

Dual-Transformer Cross-Attention Based Multimodal Emotion Recognition

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Abstract—Computer recognition of human emotion from audiovisual signals is a problem that sits at the intersection of speech processing, computer vision, and affective computing. Despite growing research on the topic, most published systems are built and tested under clean laboratory conditions, which limits how well they work in real world scenarios. This paper proposes a dual-transformer framework for multimodal emotion recognition. Two pre-trained encoders — HuBERT [1] for speech and ViViT [2] for video — process their respective streams independently, then exchange information through a bidirectional cross-modal attention mechanism. Their 768-dimensional outputs are concatenated into a 1536-dimensional joint embedding. A post-training feature selection stage using the Walrus Optimizer (WO) [3] reduces this to 765 active dimensions via a binary mask; the model is then retrained on the masked representation. Audio training data from RAVDESS [4] is expanded from 1,440 to 10,688 samples through five augmentation transforms. On the clean test set, the fusion system achieves 0.5333 accuracy and weighted F1 of 0.5085 before feature selection; after WO retraining these improve to 0.5667 and 0.5566 respectively. The HuBERT-only audio baseline, evaluated under MUSAN [5] environmental noise, achieves weighted F1 of 0.73 at 15 dB SNR and 0.56 at 0 dB SNR. Robustness evaluation of the full fusion model under noise is left for future work.

Index Terms—multimodal emotion recognition; HuBERT; ViViT; cross-modal attention; Walrus Optimizer; RAVDESS; MUSAN; audiovisual fusion; transformer

I. INTRODUCTION

Emotion plays a major role in the way people communicate and goes well beyond the literal content of speech. Prosodic variation, facial movement, and body posture collectively signal affective state, and human listeners integrate these cues almost effortlessly. Building systems that can do something similar — automatically and reliably — has practical value across a wide range of industries. Speech Emotion Recognition (SER) focuses specifically on the acoustic signal: features like pitch contour, speaking rate, and energy distribution carry emotional information that a model can, in principle, learn to decode. Audiovisual Emotion Recognition (AVER) extends this by adding the visual channel, which captures facial muscle dynamics and micro-expressions that are largely independent of what is being said. Research has shown that visual cues carry a larger share of emotional information during face-to-face communication than the acoustic signal alone [9], so combining both modalities is not merely additive — it introduces a genuinely different kind of evidence.

There are many practical uses of this kind of model. Clinical monitoring applications can track affect in patients with depression or autism spectrum disorder without requiring intrusive physiological sensors. Driver assistance systems can use emotional state as a proxy for fatigue or distraction. Call centres have obvious interest in recognising when a customer is frustrated. In all these contexts, the systems need to work in environments where noise, occlusion, and signal dropout are routine occurrences. That is where the gap between existing benchmarks and real-world applicability becomes most visible.

The dominant approach has shifted substantially over the past decade. Early work relied on handcrafted features — MFCCs, pitch statistics, energy contours — fed into SVMs or hidden Markov models. These systems performed reasonably well on controlled datasets but struggled with speaker variability and background noise. Deep learning changed the picture by learning representations directly from data: CNNs extracted spatial features from spectrograms, LSTMs modelled temporal dynamics, and eventually large-scale pre-trained transformers like wav2vec 2.0 [6] and HuBERT [1] brought self-supervised speech representations that outperformed all prior approaches on standard benchmarks. On the visual side, Video Vision Transformers [2] have enabled effective spatiotemporal feature extraction from video without the manual engineering that earlier face-analysis pipelines required.

What has not kept pace is evaluation in real world conditions. The vast majority of published AVER results are reported on clean data recorded in controlled environments, often with both modalities intact and uncompressed. When one channel degrades and the fusion strategy is fixed — as it is in most concatenation or late-fusion schemes — the corrupted channel can actively hurt performance rather than help it. This compound degradation scenario, where the system needs to dynamically rely more on whichever modality is currently trustworthy, is almost entirely absent from the experimental literature.

This paper proposes a dual-transformer AVER architecture designed with that limitation in mind. Independent HuBERT and ViViT encoders process audio and video streams, and a bidirectional cross-modal attention module enables each modality to condition its representation on the other. A post-training feature selection stage via the Walrus Optimizer [3] removes redundant dimensions from the joint embedding

before final classification. The audio-only HuBERT baseline is evaluated systematically under four levels of MUSAN environmental noise, establishing a concrete robustness floor. Clean-condition evaluation of the full fusion model is reported with and without the WO stage.

The main contributions are:

- A dual-transformer audiovisual emotion recognition system combining pre-trained HuBERT and ViViT encoders via bidirectional cross-modal attention, where each modality actively queries the other’s representation.
- An audio augmentation pipeline — pitch shifting, gain perturbation, additive Gaussian noise, room reverberation, and composed pitch-plus-reverb transforms — that expands 1,440 synchronised RAVDESS utterances to 10,688 training samples while preserving emotional label integrity.
- Hard post-training feature selection via the Walrus Optimizer, which identifies a binary mask over the 1536-dimensional joint embedding and retraining the classifier on the 765 retained dimensions, yielding a measurable improvement in both accuracy and F1.
- Systematic noise robustness evaluation of the HuBERT baseline at SNR levels of 15, 10, 5, and 0 dB using real environmental noise from MUSAN.

II. BACKGROUND

A. Emotion Representation

Describing emotion computationally is harder than it looks. Two broad schools of thought exist: one treats emotion as a set of discrete, nameable states — anger, sadness, joy — while the other maps it onto continuous axes of valence, arousal, and dominance. Both have merit, but for a classification system, discrete labels are simply easier to work with. They turn the problem into a standard supervised learning task, and RAVDESS conveniently provides exactly that: eight labelled categories covering neutral, calm, happy, sad, angry, fearful, disgust, and surprised. On the sensing side, audio and video are the obvious choices for any real deployment — no wearables, no electrodes, just a microphone and a camera, both of which are already everywhere.

B. Self-Supervised Speech Representations

Getting enough labelled emotional speech data is genuinely difficult. People disagree on what “slightly sad” sounds like; professional actors produce clean but limited recordings; and collecting naturalistic data at scale raises its own privacy and consent complications. The practical workaround is self-supervised pre-training: train a model on vast amounts of unlabelled audio first, then fine-tune it on the small labelled set you actually have. HuBERT does this by clustering raw audio frames into discrete pseudo-labels and training the network to predict those labels for masked segments — the result is a model that learns phonetic and prosodic structure on its own, without ever seeing an emotion annotation. Wav2vec 2.0 takes a slightly different route using contrastive learning, but

the underlying idea is the same. Both have become the go-to starting point for speech emotion tasks precisely because they bring so much prior knowledge to the table before any emotion-specific training begins.

C. Video Transformers for Facial Dynamics

It turns out people read faces more than they listen to voices when judging how someone feels. That makes video not just a nice-to-have but an insightful evidence. The challenge is that video is expensive — full spatiotemporal attention over every pixel of every frame is computationally heavy. ViViT gets around this by splitting attention into two separate passes: one across spatial patches within a frame, and one across the same positions over time. You get strong spatiotemporal representations without paying the full cost of joint attention. Crucially, pre-training on large action recognition datasets means ViViT already understands things like head pose shifts, eyebrow raises, and mouth movements before it ever sees an emotion label — which is exactly the kind of prior knowledge that helps on a small corpus like RAVDESS.

D. Cross-Modal Attention

The naive way to combine audio and video is to run both models, concatenate whatever comes out, and hope the classifier figures out the rest. That works well enough under clean conditions, but it breaks badly when one channel degrades — the classifier has no way of knowing that half its input is now mostly noise. Cross-modal attention offers a more principled alternative instead of merging fixed vectors, each modality actively *queries* the other, retrieving only the parts of the other stream that are actually relevant to it:

$$\mathbf{z}_{a \leftarrow v} = \text{softmax}\left(\frac{Q_a K_v^\top}{\sqrt{d_k}}\right) V_v \quad (1)$$

Run bidirectionally, this means audio can reach into the visual stream for facial cues when it needs them, and video can do the same with audio. Neither modality is passive — both are actively compensating for whatever the other might be missing.

E. Swarm-Based Feature Selection

Stacking two 768-dimensional encoders gives a 1536-dimensional joint embedding, and not all of it is useful. Some dimensions will carry redundant information that both modalities already captured independently; others will be statistical noise that happens to fit the training set. With only 120 validation samples, there is no realistic way for the classifier to learn to ignore bad dimensions on its own. The Walrus Optimizer approaches this differently: it maintains a population of binary masks over the feature space and evolves them over generations, balancing broad exploration early on with focused exploitation later through a layered danger-safety signalling mechanism. The result is a hard selection of which dimensions survive into the classifier — no soft weighting, no gradual shrinkage, just a clean binary decision for each feature. We apply it here to trim the 1536-dimensional embedding down to the dimensions that actually earn their place.

III. DATASET AND PREPROCESSING

A. RAVDESS

Finding a dataset that gives you clean, synchronised audio and video for the same emotional performance is surprisingly difficult. Most corpora either separate the two modalities across different recording sessions, or sacrifice audio quality for scale, or both. RAVDESS is one of the few that gets this right: 24 professional actors — equal split of male and female — each recorded under controlled studio conditions producing the same eight emotional categories we work with: neutral, calm, happy, sad, angry, fearful, disgust, and surprised. The total comes to 1,440 utterances, all of them genuine studio quality with no background noise baked in.

RAVDESS ships as three separate subsets: audio-only, video-only, and combined audio-video. The audio-only and video-only files come from different recording takes, meaning they are not temporally aligned with each other. If you naively pair audio from one take with video from another, you introduce a subtle but real problem — the model ends up learning cross-modal correlations that do not actually exist in nature, because the face and voice it sees together never actually belonged to the same moment. To avoid this entirely, we extract both the audio track and the video frames directly from the combined audio-video files. That way, synchronisation is guaranteed at the sample level — the face and the voice always come from the exact same performance.

B. Data Split and Augmentation

We partition the corpus into three non-overlapping sets, summarised in Table I. The validation and test sets — 120 utterances each — are drawn exclusively from the original, unmodified recordings. This is a deliberate choice: if you let augmented samples into evaluation, you end up measuring how well your augmentation pipeline works rather than how well your model generalises. Clean evaluation sets keep the numbers honest.

1,440 utterances spread across eight emotion classes and two encoders is not a lot to work with, so we expand the audio training data through five transforms applied to the training partition’s audio streams. Specifically: light and heavy pitch shifting (plus a micro-pitch variant that sits between the two), gain and volume perturbation, additive Gaussian noise at randomised SNR, room reverberation simulated via a room-impulse-response approach, and a composed transform that chains pitch shift and reverb together. These five transforms bring the audio training corpus from 1,440 to 10,688 samples.

Video frames are intentionally left unaugmented. The reason is straightforward: spatial transforms like flipping or cropping would break the spatial correspondence between the audio and visual streams that we worked hard to preserve through synchronised extraction. Beyond that, all five audio transforms share a key property — they change the absolute acoustic character of the signal (its pitch, energy, or spectral colouring) without touching the relative prosodic patterns that actually carry emotional information. The label stays valid across

every augmented variant because the emotional content of the utterance remains intact even when it sounds like it was recorded in a different room or at a slightly different pitch.

TABLE I
DATASET CONFIGURATION

Split	Source	Samples
Train	Augmented audio (5 transforms)	10,688
Validation	Pure RAVDESS	120
Test	Pure RAVDESS	120

A. Audio Encoder: HuBERT

The audio side of the system is built on HuBERT-base, and the choice is deliberate. Most emotion recognition work that uses raw waveforms still relies on hand-engineered features like MFCCs as an intermediate step. HuBERT skips that entirely — its convolutional front-end operates directly on the raw signal, producing frame-level acoustic representations at 20 ms intervals, which then pass through a 12-layer transformer that builds up temporal context across the full utterance via self-attention. The result is a single utterance-level vector $\mathbf{z}_a \in \mathbb{R}^{768}$ obtained by mean-pooling across the time axis.

Fine-tuning on a corpus of roughly 10,000 samples requires some care about what you actually let the gradient touch. The convolutional stack does low-level signal processing that generalises well across tasks — updating it risks degrading filter banks that took orders of magnitude more data to learn in the first place. So we freeze the entire CNN front-end. Within the transformer, the lower layers encode foundational acoustic properties: pitch contour, phoneme boundaries, intensity variation. These are kept frozen too. Only the upper transformer layers are left trainable, which gives them room to specialise toward the prosodic patterns that distinguish emotional categories in RAVDESS without disturbing everything underneath.

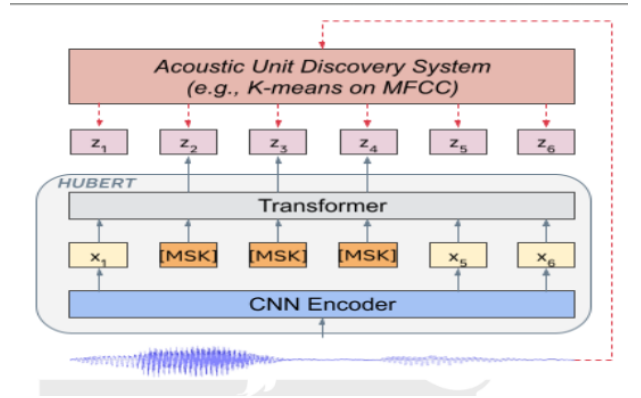


Fig. 1. HuBERT architecture. The CNN front-end processes the raw waveform directly; the transformer captures long-range temporal dependencies via self-attention. Pre-training learns to predict discrete cluster IDs for masked frame spans.

B. Video Encoder: ViViT

Faces are not static images. A raised eyebrow held for half a second means something different from one that flickers and

disappears, and capturing that distinction requires modelling both spatial structure and temporal dynamics together. ViViT-base does this by first carving each video frame into fixed-size spatial patches, projecting them into embeddings, and then running two separate transformer passes: one that attends across patches within each frame and one that attends across the same spatial positions over time. Factorising the attention this way keeps compute manageable while still giving the model a genuine sense of how the face evolves from frame to frame. The encoder runs at 32 frames per second and produces a single clip-level embedding $\mathbf{z}_v \in \mathbb{R}^{768}$.

The same partial freezing logic from the audio encoder applies here. ViViT’s lower layers — pre-trained on Kinetics — already understand things like head pose dynamics, eye movements, and the gross spatial layout of a human face. We freeze them and let only the upper layers adapt, which is enough to shift the model’s attention toward the subtler expression differences that separate.

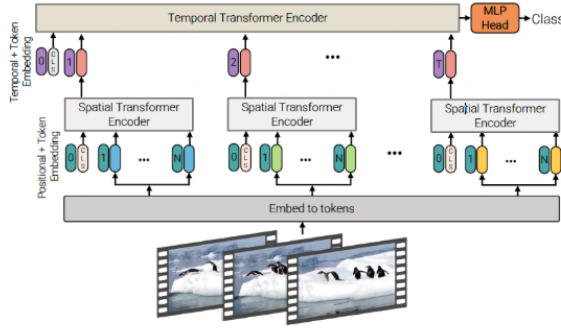


Fig. 2. ViViT architecture. Spatial and temporal transformer encoders process tokenised video patches in separate passes before final aggregation.

C. Bidirectional Cross-Modal Attention

Once HuBERT and ViViT have each produced their 768-dimensional embeddings, the question is how to combine them. The simplest answer — concatenate and classify — works fine when both channels are clean and reliable. But it falls apart the moment one of them degrades, because the classifier has no mechanism to know that half its input has become noise. We need something that lets each modality actively reach into the other and retrieve whatever is useful, rather than just passively receiving a fixed-size vector.

Cross-modal attention provides exactly that. Queries from the audio stream are matched against keys and values from the visual stream — so audio can pull out the facial cues most corresponding to what it is hearing:

$$\mathbf{z}_{a \leftarrow v} = \text{softmax} \left(\frac{Q_a K_v^\top}{\sqrt{d_k}} \right) V_v \quad (2)$$

We run this in both directions simultaneously. The visual stream queries the audio stream by the same mechanism:

$$\mathbf{z}_{v \leftarrow a} = \text{softmax} \left(\frac{Q_v K_a^\top}{\sqrt{d_k}} \right) V_a \quad (3)$$

Both operations use $d_k = 768$, and each produces an output in $\mathbb{R}^{768}$. The two cross-attended representations are then concatenated into the joint embedding:

$$\mathbf{z}_{\text{joint}} = [\mathbf{z}_{a \leftarrow v} \parallel \mathbf{z}_{v \leftarrow a}] \in \mathbb{R}^{1536} \quad (4)$$

The bidirectionality matters. If audio is corrupted by noise, the audio queries can still retrieve facial cues from the visual keys and partially recover what the acoustic signal has lost. The visual queries do the same in reverse — picking up residual prosodic signal from audio even when that channel is weak. Running both directions in parallel rather than sequentially keeps the added computational cost modest.

D. Walrus Optimizer Feature Selection

A 1536-dimensional embedding formed by concatenating two independently trained encoders usually carries eak features. Some dimensions will encode the same information twice — once from audio, once from video — and contribute nothing. Others will be irrelevant: dimensions that happened to correlate with labels on the training set but carry no genuine signal. On a validation set of only 120 samples, the MLP classifier cannot reliably learn to ignore these on its own — there simply is not enough signal to guide that kind of implicit regularisation.

The Walrus Optimizer takes a more direct approach. It maintains a population of candidate binary masks over all 1536 dimensions and evolves them over successive iterations, with each mask scored by its downstream classification performance on the validation set. A danger signal S_{danger} governs whether the population spreads out to explore new regions of the mask space or converges on the current best solutions. Within the exploitation phase, a safety signal S_{safety} further determines whether the population enters a reproduction mode — redistributing male walrus positions via Halton sequences and updating female and juvenile positions via movement equations — or a foraging mode that adjusts positions based on the current danger level. The search continues until a maximum iteration count T is reached.

Applied to our 1536-dimensional embedding, the WO identifies 765 active dimensions and zeros out the remaining 771. Critically, we do not simply apply the mask at inference time and call it done. The model is retrained from the masked checkpoint — the MLP classifier needs to redistribute its weights across a structurally different input, and the smoother post-retrain validation curves we observe (Section VII) confirm that this retraining step provides a better model

E. Classification Head

The final classification head is intentionally lightweight. The selected feature vector passes through layer normalisation — which stabilises the input distribution after the binary masking step — then through a single hidden layer with dropout regularisation and GELU activation, before a linear projection to eight class logits:

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	HuBERT Baseline	Fusion Model
Optimizer	AdamW	AdamW
Learning rate	5×10^{-6}	8×10^{-6}
Weight decay	1×10^{-4}	1.1×10^{-4}
Batch size	40	30
Epochs	30	40
Loss	Cross-Entropy	Cross-Entropy
Emotion classes	8	8
Training samples	10,688	10,688
Validation samples	120	120

$$\mathbf{h}_1 = \text{LayerNorm}(\mathbf{z}_{\text{sel}}) \in \mathbb{R}^{765} \quad (5)$$

$$\mathbf{h}_2 = \text{GELU}(W_1 \cdot \text{Dropout}(\mathbf{h}_1) + \mathbf{b}_1) \in \mathbb{R}^{384} \quad (6)$$

$$\hat{\mathbf{y}} = W_2 \cdot \mathbf{h}_2 + \mathbf{b}_2 \in \mathbb{R}^8 \quad (7)$$

Compressing from 765 to 384 dimensions before the output layer halves the parameter count of W_2 , which is a worthwhile tradeoff given how few test samples are available to evaluate on. The whole head is simple enough that it cannot overfit.

II. TRAINING SETUP

A. HuBERT Baseline

The audio-only baseline is trained with AdamW [11] at a learning rate of 5×10^{-6} , weight decay 10^{-4} , batch size 40, and cross-entropy loss over 30 epochs. The classification head maps the 768-dimensional mean-pooled HuBERT output directly to eight classes through the same LayerNorm–Dropout–Linear–GELU–Linear sequence, without any visual encoder or cross-attention.

B. Fusion Model

The full dual-transformer system uses AdamW at 8×10^{-6} learning rate, weight decay 1.1×10^{-4} , batch size 30, and cross-entropy loss over 40 epochs. The slightly different hyperparameters reflect the larger parameter count of the joint system: the lower learning rate reduces the risk of the cross-attention weights overwriting the pre-trained encoder representations during the early training steps, and the higher epoch count gives the cross-attention module more iterations to settle. For both models, the checkpoint with the lowest validation loss is selected for evaluation.

III. ROBUSTNESS EVALUATION PROTOCOL

For the audio-only baseline, environmental noise is sourced from MUSAN [5], which provides real-world recordings of babble, music, and ambient background sounds. A noise segment is randomly selected from MUSAN and mixed with each clean test utterance at a target signal-to-noise ratio. Four levels are tested:

- **15 dB (mild):** The speech signal clearly dominates. Pitch contour and energy envelope are largely recoverable.

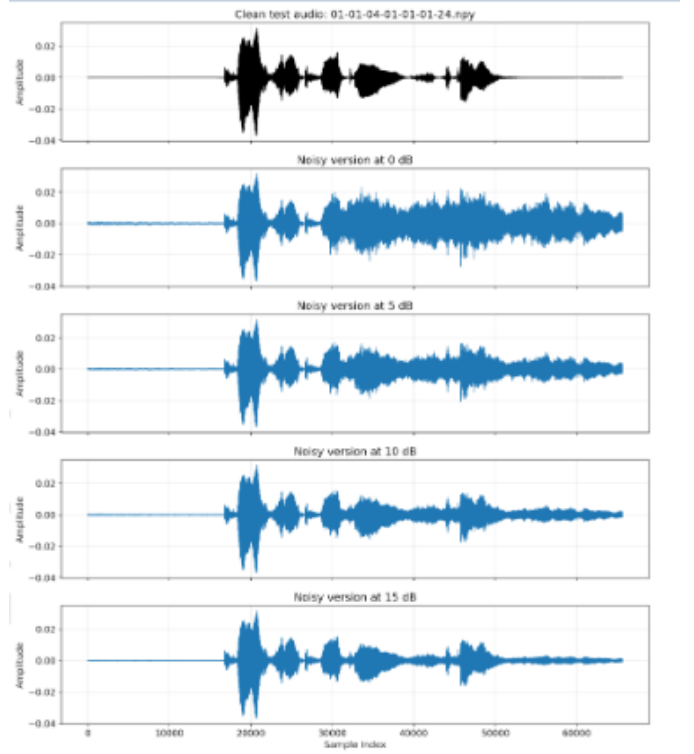


Fig. 3. A single RAVDESS test utterance under clean conditions and at 0, 5, 10, and 15 dB SNR. Speech structure is progressively masked as noise increases; at 0 dB the original signal is almost completely submerged.

- **10 dB (moderate):** Noise becomes perceptible and starts interfering with lower-energy regions of the utterance.
- **5 dB (severe):** Substantial spectral masking occurs. The model must rely heavily on pre-trained prosodic priors.
- **0 dB (extreme):** Noise power equals signal power. Most acoustic emotion cues are unreliable.

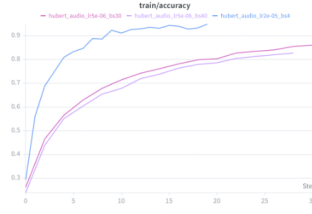
Noise is applied only to the audio channel; the video stream remains clean throughout. This models a practically common scenario — outdoor or in-car noise degrading the microphone while the camera is unaffected — and is exactly the condition where the visual stream of the dual-transformer system should provide the most benefit. Noise robustness evaluation of the full fusion model is deferred to future work; the clean-condition fusion results and the baseline noise results together establish the relevant reference points.

IV. RESULTS AND DISCUSSION

A. HuBERT Baseline: Training

Fig. 4 shows training and validation curves for three HuBERT configurations. The run with $\text{lr} = 5 \times 10^{-6}$, batch size 40 converges fastest, reaching training accuracy above 0.90 within the first ten epochs. Validation loss for the two lower-learning-rate runs (5×10^{-6} , batch 30 and 5×10^{-6} , batch 40) decreases steadily and stabilises around 0.6, indicating reasonable generalisation. The higher-learning-rate configuration (2×10^{-5} , batch 4) shows substantially more oscillation in validation loss, which is expected: the larger gradient steps

TRAINING:



EVALUATION:



Fig. 4. HuBERT baseline: training accuracy (left) and validation loss (right) for three hyperparameter configurations over 30 epochs.

TABLE III

HUBERT AUDIO-ONLY BASELINE: PERFORMANCE UNDER MUSAN NOISE

SNR (dB)	Accuracy	Macro F1	Weighted F1
15 (mild)	0.75	0.73	0.73
10 (moderate)	0.73	0.71	0.70
5 (severe)	0.68	0.67	0.66
0 (extreme)	0.58	0.56	0.56

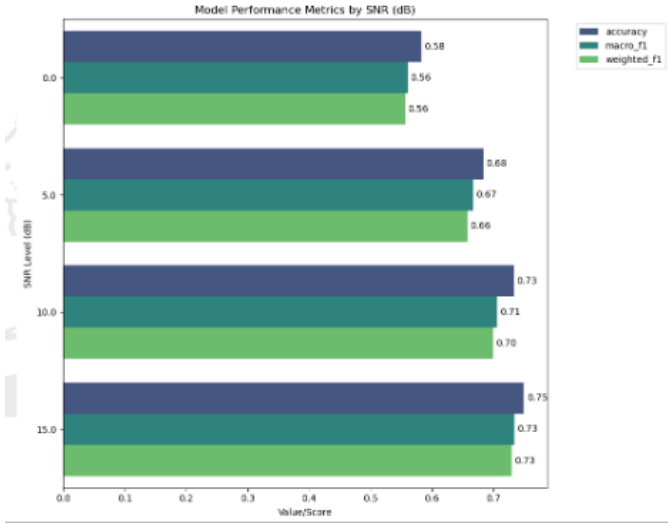


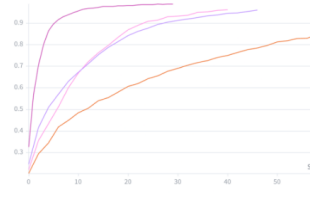
Fig. 5. HuBERT baseline accuracy, macro F1, and weighted F1 across four MUSAN SNR levels. The sharpest drop occurs between 5 dB and 0 dB.

are less stable on a validation set of only 120 samples, where a few misclassified samples produce noticeable loss swings. The selected checkpoint is $lr = 5 \times 10^{-6}$, $bs = 40$.

B. HuBERT Baseline: Noise Robustness

Table III and Fig. 5 summarise the results. At 15 dB, the system achieves 0.75 accuracy and 0.73 weighted F1. Degradation from 15 to 5 dB is gradual: only 7 percentage points of accuracy are lost across that range, which likely reflects the benefit of the Gaussian noise augmentation applied during training. The model has seen noisy variants of training samples and has developed some tolerance for acoustic corruption, at moderate levels.

TRAINING:

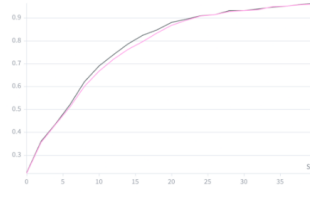


EVALUATION:



Fig. 6. Fusion model training accuracy (left) and validation loss (right) for four hyperparameter configurations over 40 epochs.

TRAINING:



EVALUATION:

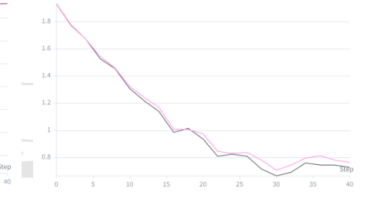


Fig. 7. Optimised retraining curves after the Walrus binary mask is applied. Training accuracy (left) and validation loss (right) over 40 epochs. Validation loss shows noticeably smoother convergence than the pre-mask training run.

The 0 dB case is notably worse: accuracy drops to 0.58 and weighted F1 to 0.56. At this SNR, noise power equals signal power and the spectral content that encodes prosodic emotion cues is heavily corrupted. It is worth noting that 0.58 is still well above the 0.125 expected by chance across eight classes, so the pre-trained HuBERT representations are not entirely overwhelmed even under extreme noise. But the gap between 5 dB (0.68) and 0 dB (0.58) is twice as large as any adjacent gap in the table, suggesting that 0 dB represents something close to a practical operating limit for audio-only SER under unconstrained noise.

C. Fusion Model: Training

Fig. 6 shows training and validation dynamics for four fusion configurations. The $lr = 8 \times 10^{-6}$, $bs = 30$ run achieves the highest training accuracy, approaching 0.95 by epoch 40, and its validation loss descends most smoothly from an initial value near 2.0 to roughly 0.65 by the end of training. The lower-learning-rate configurations ($lr = 2 \times 10^{-6}$) converge more slowly and do not fully utilise the 40-epoch budget, suggesting the cross-attention weights require more gradient signal per step than those runs provide. The highest learning rate run ($lr = 1 \times 10^{-5}$, $bs = 10$) shows more erratic validation behaviour, consistent with instability from larger updates on a small validation set.

D. Post-WO Retraining

Fig. 7 shows retraining curves after the 765-feature binary mask is applied. The training accuracy trajectory is nearly identical to the pre-mask run — the encoder representations and cross-attention weights were not reset, so the model starts

TABLE IV
FUSION MODEL: CLEAN TEST SET PERFORMANCE BEFORE AND AFTER WO

System	Accuracy	Macro F1	Weighted F1
Fusion, pre-WO (1536-dim)	0.5333	0.5080	0.5085
Fusion, post-WO (765-dim)	0.5667	0.5531	0.5566
Gain	+0.0334	+0.0451	+0.0481

from a strong initialisation. What changes is the validation loss curve: it decreases more smoothly and stabilises at a lower value than before masking, bottoming out near 0.7 by epoch 25 and staying there. The pre-mask curve showed more oscillation in the same region. Removing 771 dimensions appears to reduce the effective complexity enough that the small validation set can provide a more consistent gradient signal.

E. Clean Test Performance

Table IV reports clean-condition test performance. Before WO, the system achieves 0.5333 accuracy and 0.5085 weighted F1. After the WO identifies 765 active features and the model is retrained on the masked representation, accuracy rises to 0.5667 and weighted F1 to 0.5566. The gains — 3.3 points in accuracy, 4.5 in macro F1, 4.8 in weighted F1 — are modest but consistent across all three metrics, which suggests they reflect a genuine improvement rather than noise on the small test set. Average inference latency also drops from 8.79 ms to 7.81 ms per embedding, a small but useful side effect of the reduced input dimensionality.

Calm improves from F1 0.7273 to 0.7647, driven by higher recall (0.75 to 0.8125), meaning the post-WO model misclassifies fewer calm samples as neutral. Angry improves from F1 0.6000 to 0.6667 through better precision. Disgust and Surprised both improve slightly. Happy is the weakest category in both configurations — F1 0.1111 before WO and 0.2609 after — which is not surprising given how acoustically and visually similar happy and surprised expressions can be in acted datasets like RAVDESS. The confusion matrices in Figs. 8 and 9 make this pattern visible: Happy samples consistently scatter across Sad, Angry, and Surprised columns in both cases, and the improvement after WO comes mainly from a slight increase in correctly identified Happy samples.

F. Discussion

The HuBERT-only baseline outperforms the fusion system on clean data by a significant margin (0.75 vs. 0.5667). This gap deserves some understanding rather than being treated as a failure of fusion. The HuBERT baseline trains a single encoder and a two-layer classifier on 10,688 audio samples; the fusion system simultaneously optimises two encoders, a bidirectional cross-attention module, and the same classifier architecture. The additional parameters introduced by cross-attention require more data to converge reliably, and the RAVDESS corpus — even after augmentation — is small for

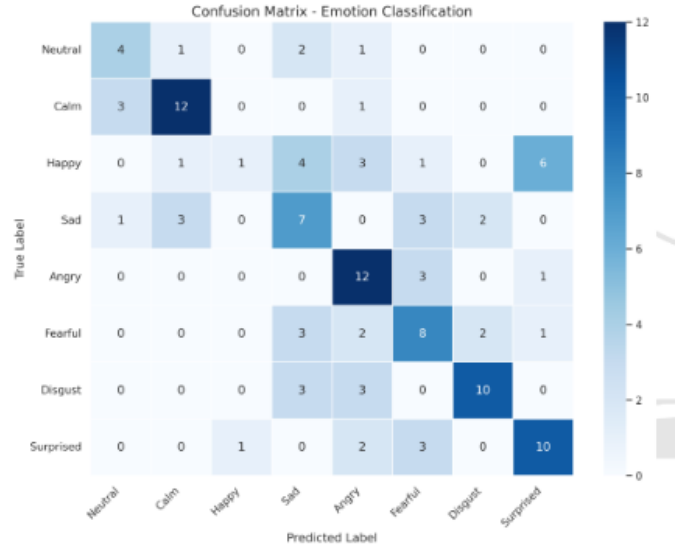


Fig. 8. Confusion matrix for the fusion model on the clean test set *before* Walrus Optimizer feature selection (1536-dim input, accuracy 0.5333).

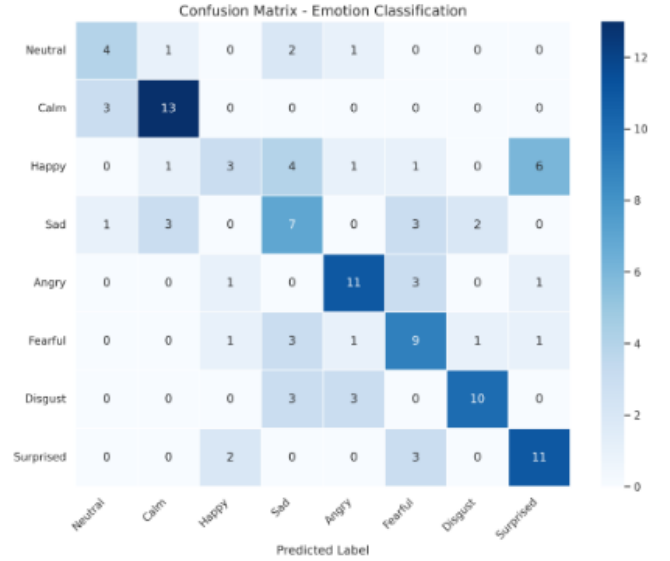


Fig. 9. Confusion matrix for the fusion model on the clean test set *after* Walrus Optimizer feature selection (765 active features, accuracy 0.5667).

a dual-encoder system. The WO gain of 3.3 accuracy points suggests the joint embedding does contain multimodal signal, but extracting it fully from 10,688 samples is difficult.

Under noisy conditions, the performance of the audio-only HuBERT baseline decreases. At 0 dB SNR the HuBERT baseline drops to 0.58. In such scenarios, the proposed multimodal fusion framework is expected to leverage visual information to compensate for degraded audio signals and improve overall emotion recognition performance. Evaluating the robustness of the fusion architecture under varying noise levels remains an important direction for future work.

TABLE V
PER-CLASS F1 SCORES: FUSION MODEL BEFORE AND AFTER WO

Emotion	Pre-WO F1	Post-WO F1
Neutral	0.5000	0.5000
Calm	0.7273	0.7647
Happy	0.1111	0.2609
Sad	0.4000	0.4000
Angry	0.6000	0.6667
Fearful	0.4706	0.5143
Disgust	0.6667	0.6897
Surprised	0.5882	0.6286
Weighted avg	0.5085	0.5566

VIII. CONCLUSION

This paper presented a dual-transformer audiovisual emotion recognition framework combining HuBERT and ViViT encoders connected through bidirectional cross-modal attention. Each modality actively queries the other’s representation, producing a 1536-dimensional joint embedding that carries genuine multimodal signal. Post-training feature selection via the Walrus Optimizer trims this to 765 active dimensions and retrains the classifier, pushing accuracy from 0.5333 to 0.5667 and weighted F1 from 0.5085 to 0.5566 on the clean test set — consistent gains across all eight emotion classes with a small reduction in inference latency as a bonus.

The HuBERT audio baseline holds up well under mild noise (0.73 weighted F1 at 15 dB) but degrades meaningfully at extreme noise (0.56 at 0 dB), marking a practical operating limit for audio-only systems. The fusion model’s lower clean-condition accuracy reflects the optimisation burden of a dual-encoder system on a small corpus — not a fundamental flaw in multimodal fusion.

The central question is whether the visual stream compensates for acoustic degradation under noise. Evaluating the full fusion model under MUSAN noise, extending to quality-aware cross-modal attention, and training on a larger corpus are the natural next steps toward a system that works in the real world.

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